StarDICE : instrumental flux calibration with an artificial star for SNe Ia cosmology with the Legacy Survey of Space and Time

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Abstract

Measurements of luminosity distances using SNe-Ia are essential to constrain the equation of state parameter ω [1], [2], [3]. Current measurements of SNe-Ia are limited by the statistics and the underlying systematic errors [4]. The large number of detected SNe-Ia with the upcoming Vera Rubin Observatory will address the first issue [5]. In the Photometric Calibration Working Group of the Dark Energy Science Collaboration of LSST, we propose the StarDICE experiment which will put effort toward the reduction of systematic errors. StarDICE, is a metrology experiment installed at Observatoire de Haute-Provence whose goal is to achieve millimagnitude calibration required to extract most of the statistical power of LSST. This implies knowing relative flux changes at < 0.1% on CALSPEC stars [6].

The StarDICE experiment

Method

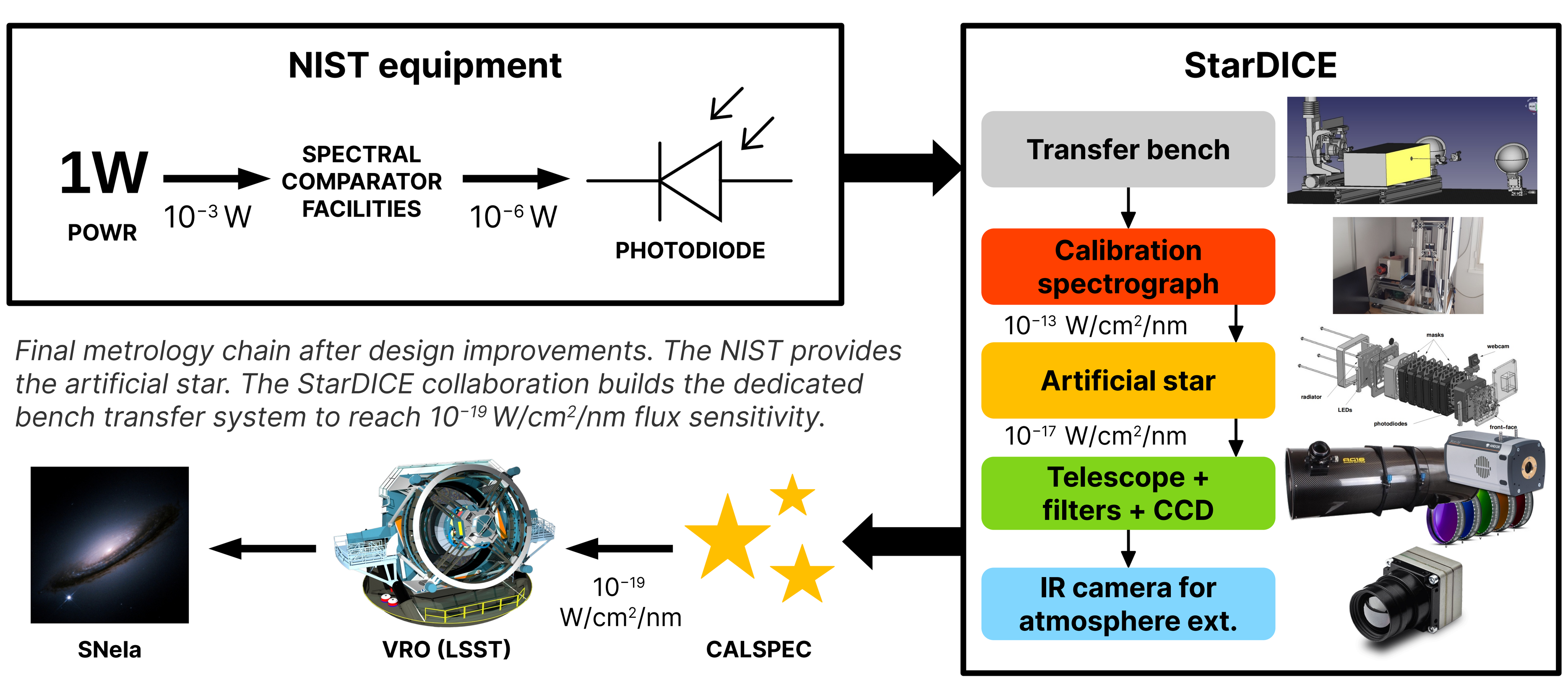
The proposed strategy is the use of a stable calibrated light-source called the artificial star, as a flux standard to calibrate the flux response of a small aperture telescope. By knowing the calibrated flux of the NIST artificial star F_T^{LED}, we can correct the response of the instrumental chain :

$$\frac{F_M^{LED}(t,\lambda)}{F_M^{STAR}(t,\lambda)} = \frac{R(h,t,\lambda)}{R(H,t,\lambda)} \cdot \frac{F_T^{LED}(\lambda)}{F_T^{STAR}(\lambda)}$$

Where R is the instrumental response with h the altitude of observation of the LED, H the altitude of observation of the star. F_T and F_M represent the theoretical and measured flux respectively. With enough observation data, we can fit F_T^{STAR} using a template to get the absolute flux of the CALSPEC star.

Prospects

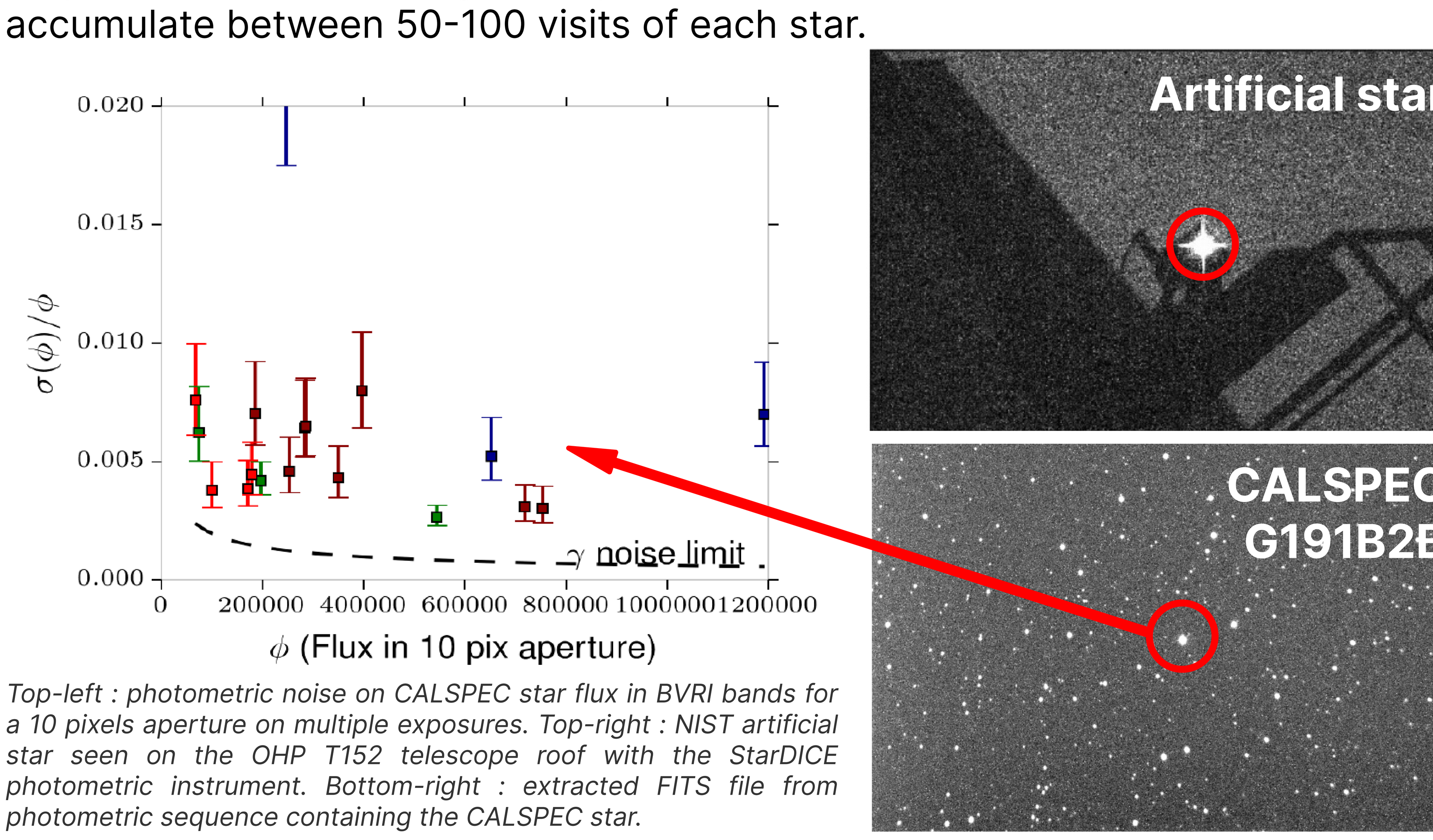
Following the proof of concept phase, design improvements have been made on every stage of the metrology chain. StarDICE commissioning and first light after upgrade at OHP is expected to occur during the first semester of 2022.



Final metrology chain after design improvements. The NIST provides the artificial star. The StarDICE collaboration builds the dedicated bench transfer system to reach 10^{-19} W/cm²/nm flux sensitivity.

Pathfinder (2016-2019)

The first version of the experiment designed to validate the concept has been installed and successfully test at the OHP by obtaining an average statistical flux uncertainty of 0.7% in each spectral band over 20 minutes of data acquisition [7]. The target of precision of 0.1% is within reach if we can accumulate between 50-100 visits of each star.



Top-left : photometric noise on CALSPEC star flux in BVRI bands for a 10 pixels aperture on multiple exposures. Top-right : NIST artificial star seen on the OHP T152 telescope roof with the StarDICE photometric instrument. Bottom-right : extracted FITS file from photometric sequence containing the CALSPEC star.

References

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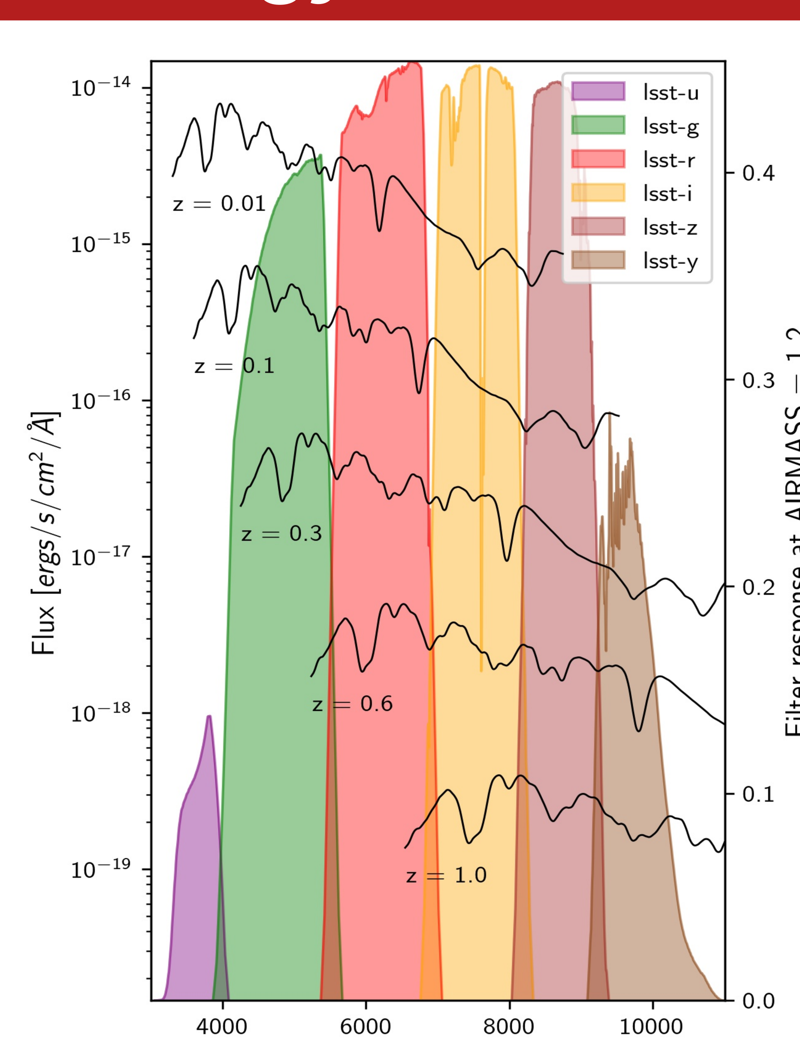
[7] Hazenberg, *Calibration photométrique des supernovae de type Ia pour la caractérisation de l'énergie noire avec l'expérience StarDICE*, PhD thesis, 2019
[8] Mandelbaum et al., *The LSST Dark Energy Science Collaboration (DESC) Science Requirements Document*, 2021

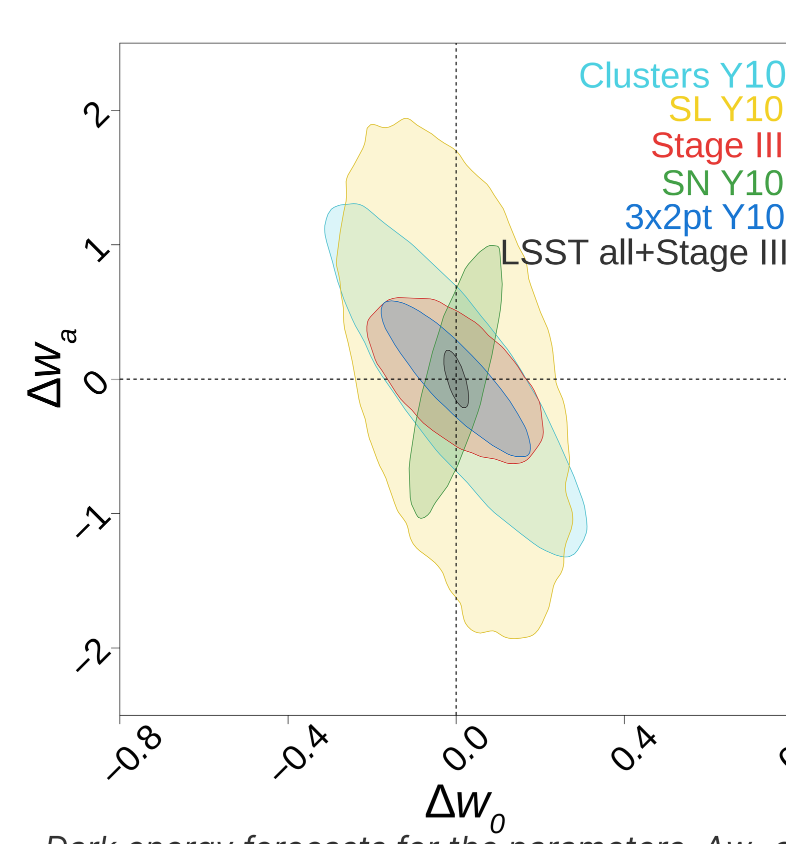
SNe-Ia cosmology with the Legacy Survey of Space and Time

SNe-Ia study with the VRO will be based on photometry of the broad-band colors (ugrizy) from either the host galaxy or the SN itself. If the intrinsic spectrum is known, the colors have the potential to accurately determine the distance of the source with a spectrophotometric model. Using this method, the LSST data will contain ~50,000 SNe-Ia per year with high precision photometry for accurate light curve fitting and a mean redshift of z~0.5 [5]. Reaching mmag calibration is necessary in order to bring the LSST statistical power above the systematic level. This data will enable the constrain of ω_0 to ±0.05, and ω_a to ±0.10 within the redshift-dependent equation of state :

$$\omega(z) = \omega_0 + \omega_a\left(\frac{z}{1+z}\right)$$



LSST ugrizy filters responses with simulated SNe-Ia spectrum for different redshifts



Dark energy forecasts for the parameters Δw_0 and Δw_a which must be obtained with the different statement [8].
Clusters Y10 → Counting galaxy clusters
SN Y10 → Strong gravitational lensing
Stage III → Pre-LSST statements
SN Y10 → Supernovae LSST
3×2pt Y10 → Low gravitational lensing LSST
LSST all + Stage III → Total

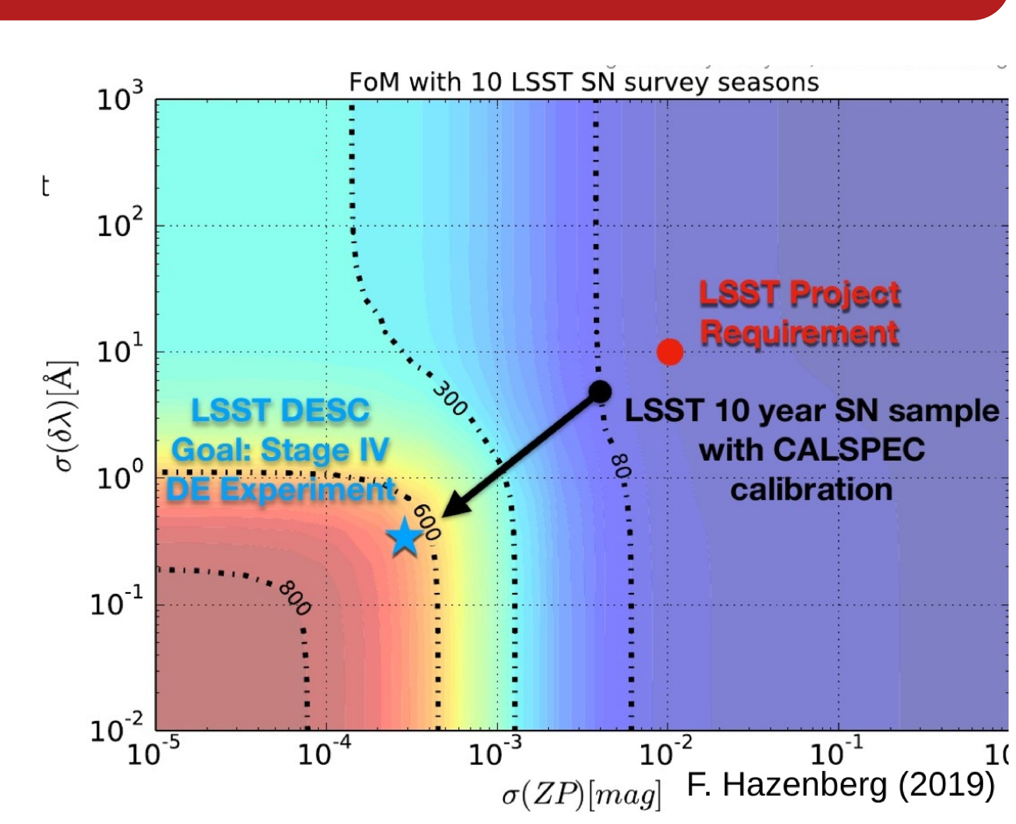


Figure of Merit (FoM) for 10-year period of LSST SN survey, defined as the inverse of the area of the ellipse of the contour at 1σ of the uncertainties in the plan (ω_0, ω_a). Different calibration stages for LSST are represented. At 1% calibration the SNe-Ia survey does not bring much to cosmology. 1 mmag precision is required to extract most of the statistical power of LSST [7].

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